

Q4: Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Introduction

In digital image processing, pixel resolution and intensity resolution are two important factors that determine the quality of a digital image. In medical imaging applications such as X-rays, CT scans, and MRI scans, both resolutions play a significant role in accurately representing anatomical structures and detecting diseases.

Pixel Resolution

Pixel resolution refers to the number of pixels used to represent an image (for example, 512×512 or 1024×1024 pixels). It determines the spatial resolution or the level of detail present in the image.

Effect on Image Representation

- Higher pixel resolution captures more image details.
- Small structures such as tiny blood vessels, fractures, or tumors become clearly visible.
- Edges appear sharper and more accurate.
- Lower pixel resolution causes images to appear blurred or pixelated, making diagnosis difficult.

Example in Medical Imaging

In an MRI scan, a high pixel resolution allows doctors to identify small tumors or tissue abnormalities that may not be visible in a low-resolution image.

Intensity Resolution

Intensity resolution refers to the number of gray levels available for representing pixel brightness. It depends on the bit depth of the image.

For example:

- 8-bit image = 256 gray levels
- 12-bit image = 4096 gray levels
- 16-bit image = 65,536 gray levels

Effect on Image Representation

- Higher intensity resolution provides smoother gray-level transitions.
- It improves image contrast and makes slight differences in tissue density visible.
- Lower intensity resolution causes loss of contrast and produces banding effects.

Example in Medical Imaging

In CT scans, high intensity resolution helps distinguish between healthy tissue and diseased tissue because even small differences in tissue density become visible.

Difference Between Pixel Resolution and Intensity Resolution

Pixel Resolution

Determines the number of pixels in an image.

Affects spatial detail and image sharpness.

Higher resolution improves visibility of small objects.

Measured in image dimensions (e.g., 512 × 512).

Intensity Resolution

Determines the number of gray levels (brightness values).

Affects image contrast and brightness accuracy.

Higher resolution improves differentiation between tissues.

Measured in bits (e.g., 8-bit, 12-bit, 16-bit).

Conclusion

Both pixel resolution and intensity resolution are essential for producing high-quality medical images. Pixel resolution improves spatial detail and image sharpness, while intensity resolution improves contrast and enables better differentiation between tissues. In medical imaging, using high pixel resolution together with high intensity resolution helps doctors detect diseases more accurately and make reliable diagnoses.